

FRANCIS PADRON Calgary, Alberta | 403-918-5507

padronfrancis1@gmail.com | [linkedin.com/in/francis-padron](https://www.linkedin.com/in/francis-padron) | padronfrancis1.github.io/portfolio

Senior Generative AI Developer | AI/ML Engineer

PROFESSIONAL PROFILE

- 10+ years of software development experience, with 2+ years focused on AI/ML and Generative AI solutions
- Built production RAG, multi-agent, and agentic AI systems using LangChain and LangGraph
- Integrated OpenAI, Azure OpenAI, AWS Bedrock (Llama 3.3), Ollama, and Hugging Face LLMs
- Vector search with Qdrant and ChromaDB, embeddings, reranking, and prompt engineering
- Cloud-native and MLOps focus - Docker, AWS ECS Fargate, CI/CD, Responsible AI guardrails, and governance

TECHNICAL SKILLS

Tools: Python, FastAPI, LangChain, LangGraph, OpenAI, Azure OpenAI, AWS Bedrock, Hugging Face, PEFT/LoRA, Unsloth, TRL, ZenML, PyTorch, Qdrant, ChromaDB, PostgreSQL, Docker, AWS ECS, .NET C#

EDUCATION

Post-Diploma Certificate, Artificial Intelligence, Southern Alberta Institute of Technology (SAIT), Awarded

Post-Diploma Certificate, Web Design Certificate, Southern Alberta Institute of Technology (SAIT), Awarded

Post-Diploma Certificate, SharePoint Specialization, INNOTECH College, Awarded

Microsoft® TS: Microsoft ASP.NET Framework 3.5 ASP.NET Application Development Awarded

Microsoft® TS: Microsoft .NET Framework - Application Development Foundation Awarded

Bachelor's degree, Information Technology, Colegio de San Juan De Letran, Awarded-World Education System (WES) Verified

PROFESSIONAL EXPERIENCE

Havenz Tech, Calgary, Alberta

April 2025 - Present

AI/ML Engineer | Generative AI & Agentic Systems

- Architected HavenzHub-AI, a multi-agent RAG platform on FastAPI + LangGraph orchestrating OpenAI, AWS Bedrock, and Ollama with cost-optimized 3-tier routing
- Designed Qdrant hybrid vector search with cross-encoder reranking, parent-child retrieval, and persistent semantic memory
- Built sub-agent architecture (Document, Business, Gmail) with HMAC-verified webhook ingestion and Responsible AI guardrails
- Delivered an AI Trading Bot using a local Qwen LLM for trade analysis on a Flask + IBKR/Kraken multi-asset architecture
- Engineered a multi-model CV pipeline (YOLOv8, Mask R-CNN, Mask2Former, RF-DETR) cutting hail-dent assessment time by 70%
- Fine-tuned LLMs (Llama 3.1 8B, Qwen2.5, SmolLM2) with SFT, DPO, and LoRA via Unsloth + TRL, orchestrated with ZenML
- Built electricity pool-price forecasting with two-stage regime-switching LightGBM and walk-forward CV on AESO data
- Established CI/CD with 3,400+ unit tests, Dockerized AWS ECS Fargate deployment, and production monitoring

Sleek Signs, Calgary, Alberta

Jun 2025 – Oct 2025

AI Engineer & Full-Stack Engineer (Contract)

- Deployed an on-premises local-LLM RAG stack (Ollama + ChromaDB) with PII and guardrail middleware for data privacy and governance
- Built an automated Google Drive ingestion pipeline (FastAPI) that syncs documents into the vector DB on schedule
- Built a Quote Agent that converts natural-language requests into priced signage quotes via LLM tool calling
- Integrated Zoho CRM as agent tools (client search, quote history, invoices) for unified CRM and pricing chat
- Built a daily feedback pipeline that retrains the routing classifier from user thumbs-up/down, improving reranking accuracy

Jim Pattison Lease, Calgary, Alberta

Dec 2021-Jul 2024

Software Developer and RPA Specialist (Automation Anywhere)

- Developed and optimized a CRM system to streamline daily car lease operations
- Customized and tested Prairie Dog CRM (.NET, C#) for the US market, ensuring high code quality
- Thrived in Agile environments, adapting quickly to fast-paced development cycles
- Handled data-driven operations with SQL & Entity Framework, improving data integrity and access
- Set up DevOps pipelines using GitHub, Azure, AppVeyor, and Octopus, enabling smooth CI/CD workflows

AECOM, Calgary, Alberta

Mar 2020-Dec 2021

Full Stack Developer/Data Management

- Technical Lead on CRM development, managing employee data, and HR workflows
- Designed and customized SQL reports, enabling data-driven decisions for stakeholders
- Maintained system stability by resolving bugs and issues in .NET (C#) applications
- Handled continuous development and deployment via Microsoft Azure for high uptime
- Built new features based on user needs, ensuring alignment with evolving requirements
- Collaborated with analysts and developers to deliver solutions that met business goals

AFTI WatchDog, Calgary, Alberta

Jan 2017-Mar 2019

Systems Developer/ Full Stack Developer

- Processed audio signal data for well monitoring and leak detection using time-series ML
- Built interactive time-series dashboards using AngularJS, JavaScript, and D3.js for real-time insights
- Developed a full-stack ASP.NET (C#) web app for efficient client data management
- Created a REST API in Python to automate and streamline billing report generation
- Designed SQL schemas, stored procedures, and views, ensuring reliable backend data handling

Cenovus Energy, Calgary, Alberta

Feb 2013-Jan 2016

Data Management/ Systems Integrator

- Managed SCADA sensor data to support reporting and operational decision-making
- Generated custom SQL reports as part of the Data Management team for data integrity
- Worked with vendors (OSIsoft PI System and PARCView), enhancing data monitoring workflows
- Delivered client reports and resolved data issues, ensuring accuracy and system uptime
- Built tools in .NET (C#) and PI SDK to detect and correct data discrepancies efficiently
- Automated Excel workflows with VB.NET, enabling faster insights for ENMAX
- Processed data from 3,000+ folders using PowerShell, streamlining large-scale analysis